AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 3 · STUDENT LESSON

Sensors Are Messengers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Read the measurement. Limit the claim.

Start with an unexplained 24, trace the garden warning, compare paired light readings, then investigate a new disagreement.

Keep one habit: Name the quantity and conditions → compare fairly → say what is still unknown.

MAIN LESSON · SECTION 1

What does a sensor reading really tell us?

> Describe a measurement, trace how it is used and compare devices fairly.

Start with a disagreement

Two devices show different numbers. Before choosing which one to trust, ask: what was measured, in which units, where and when? This week you will separate a reading, a rule and a claim about the world.

Reconnect with information loss

A record retains only the highest temperature of a day. Can it establish when that temperature occurred? Explain using two possible records.

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can describe a **measurement** using its quantity, value, **unit** and conditions.
- I can distinguish a **sensor** reading from an interpretation or decision.
- I can use repeated comparisons to investigate a **measurement** problem without inventing certainty.

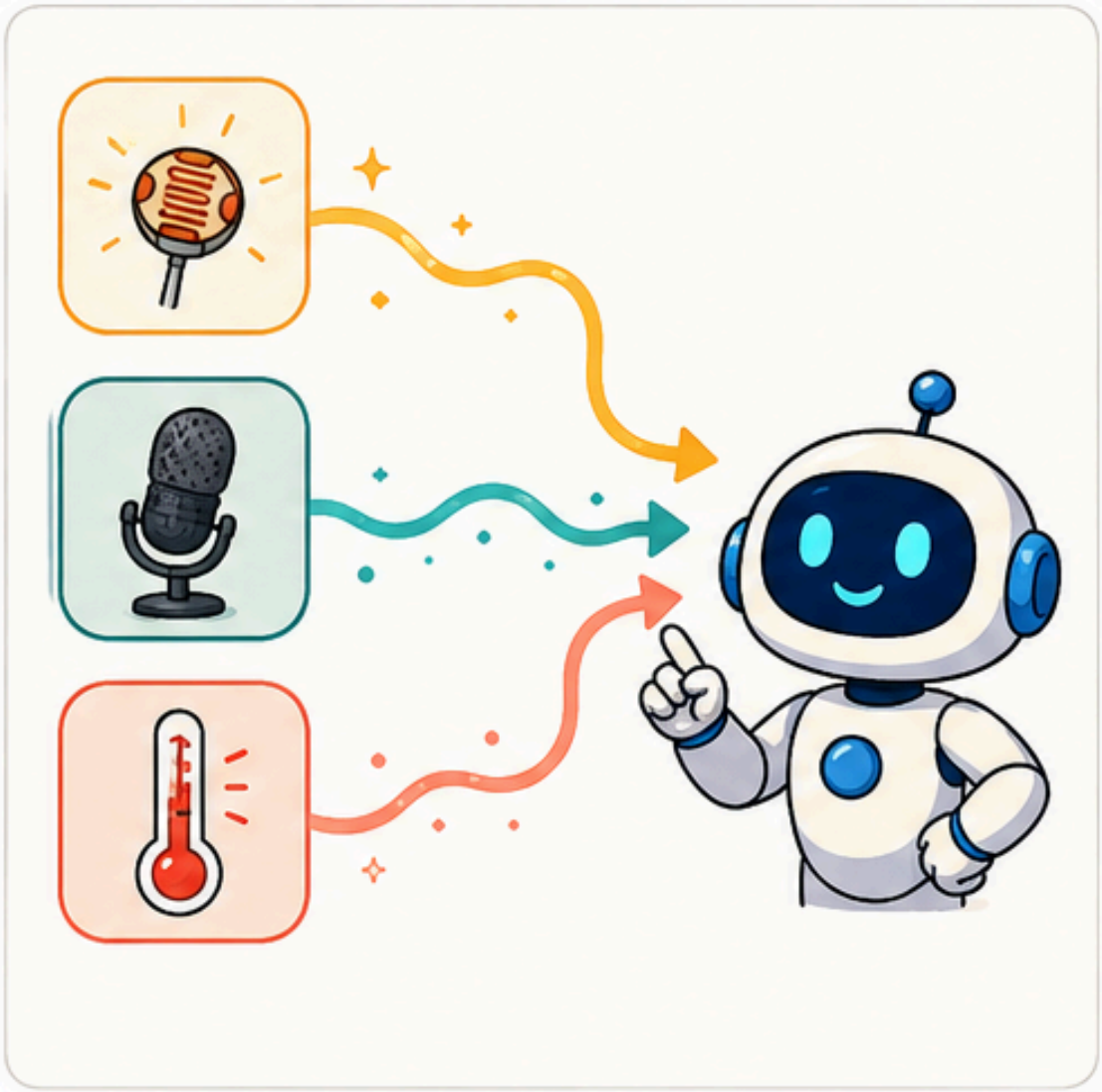
▼ Earlier wording, alternative practice and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

Sensors measure **signal**s; later **process**ing gives those **signal**s meaning.



LEARNING TARGET Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright?

Design a fair comparison without using a camera or personal **data**.

Learner name:

Readiness for the connected explanation

This week's end goal

By the end, I can compare **sensor** readings while controlling where and when they are measured.

MAIN LESSON · SECTION 2

The display says 24. Twenty-four what?

> A number needs a quantity, unit and conditions before it becomes useful evidence.

Read the garden display

A garden display shows 24. Is that warm? We cannot interpret the number until we know what was measured and the **unit**.

It might mean 24 degrees Celsius, 24 percent soil moisture according to a particular device, or 24 hours since watering. The screen alone does not supply the missing meaning.

Complete the measurement record

A useful **measurement** record names the quantity, value, **unit**, time and relevant conditions. For air temperature, the location and exposure matter: a device in direct sunshine may warm differently from one in shade.

Record the conditions rather than assuming that every nearby device measured the same situation.

A temperature is not a watering decision

Measurement is a way of representing part of the physical world. It does not capture everything about that world.

One air-temperature reading does not tell us how dry the roots are, whether a plant is diseased or whether watering is permitted.

Notice before explaining

- Name three visible details without explaining them yet.
- Make a first **prediction** and state the clue that supports it.

▼ Earlier wording, alternative practice and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A classroom weather station measures light, sound, and temperature but does not decide what those **measurement**s mean.

What do you notice?

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

MAIN LESSON · SECTION 3

Measurement → rule → human decision

> Follow three different jobs instead of treating the sensor as a gardener.

Trace the garden system

Three different stages in a garden system

INPUT	OPERATION	OUTPUT
Physical temperature at the sensor	Sensor and measuring electronics	Recorded air temperature 24 degrees C
Recorded value 24 degrees C	Written rule checks whether value exceeds 28	Warning is off
Readings and a gardener's inspection	Gardener decides what action is appropriate	An action or a decision to wait

Sensing, checking a rule and deciding what to do are separate stages. A correct rule can still receive an unsuitable measurement.

Explain what the sensor changes

The **sensor** responds to a physical quantity. Electronics convert the response into a recorded value. A later program can compare the value with a threshold.

Nothing about this path requires us to claim that the **sensor** understands plant health.

UNDER THE HOOD A **sensor** is a transducer: it converts a physical quantity into a **signal** a computing system can represent. Sensing and interpretation are separate.

A camera measures light; an image model may later infer a label.

Read the boundary carefully

The rule uses exceeds 28, so a reading of exactly 28 does not trigger it. At 29 it does.

This boundary is a property of the **written rule**; it is separate from the question of whether the measuring device is accurate enough for the intended purpose.

Explain one arrow

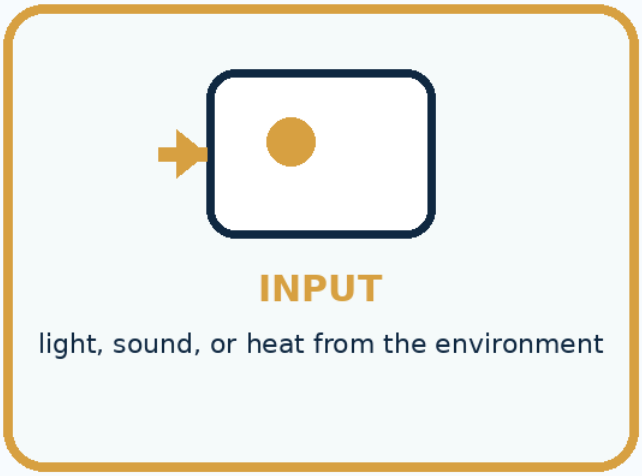
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

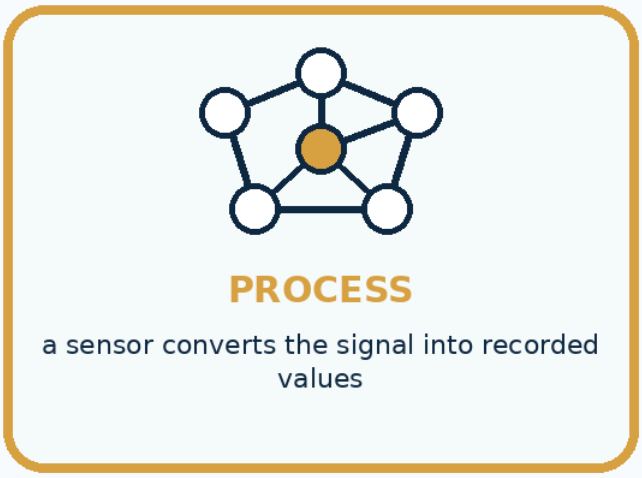
▼ **Earlier wording, alternative practice and picture transcripts**

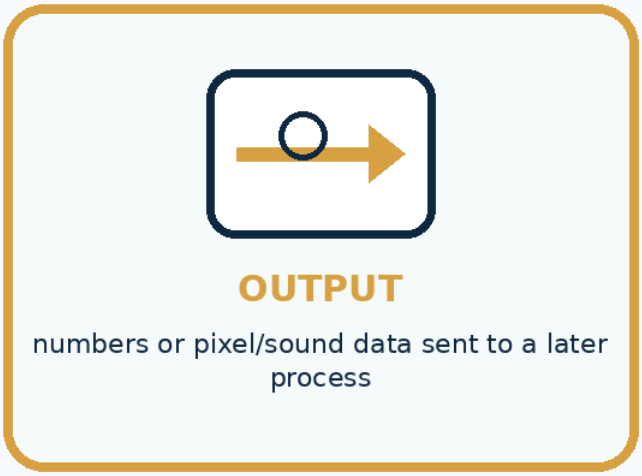
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT light, sound, or heat from the environment

PROCESS a **sensor** converts the **signal** into recorded values

OUTPUT numbers or pixel/sound **data** sent to a later **process**

CORE IDEA **Sensor**s measure **signal**s; later **process**ing gives those **signal**s meaning.

Draw a fresh example with all three stages:

A closer explanation

The sensor measures. The rule decides.

This model uses a made-up light scale, from 0 (darker) to 100 (brighter).

Light reading: 18

Rule: if reading is below 30, turn the lamp on; otherwise turn it off.

Lamp ON

Try 29, then 30. Which part measured the light, and which part chose the action?

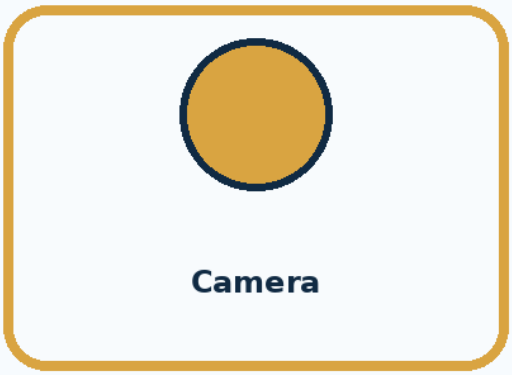
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Match the physical signal to its record

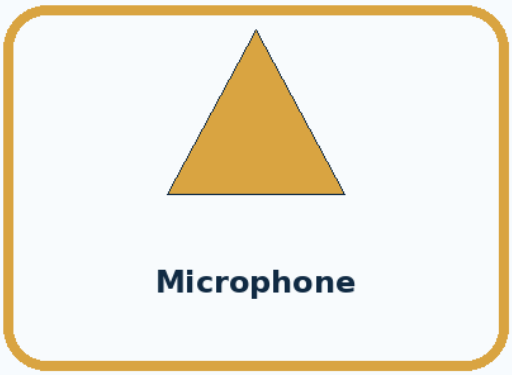
> Use each picture’s evidence to distinguish sensing from interpretation.

Four pictures with their explanations



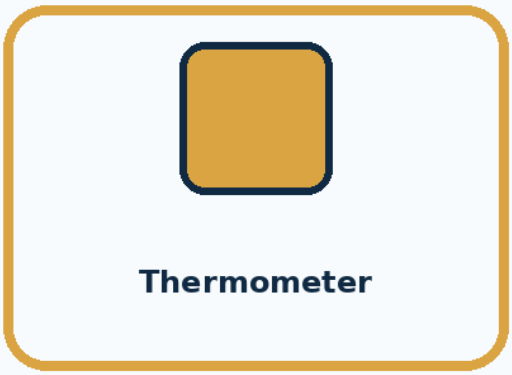
The case to investigate

CASE 1 — Camera Clue: light becomes pixel values. Meaning: **measurement**, not meaning.



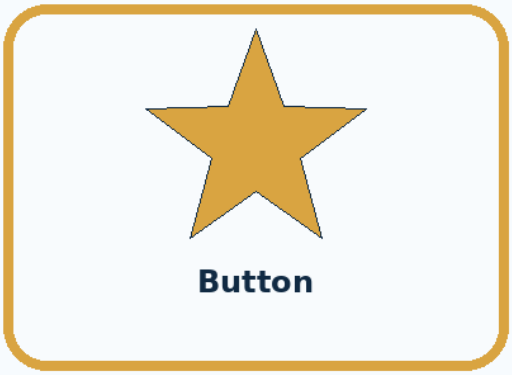
The case to investigate

CASE 2 — Microphone Clue: air-pressure changes become samples. Meaning: **measurement**, not understanding.



The case to investigate

CASE 3 — Thermometer Clue: heat becomes a temperature number. Meaning: **measurement**, not a weather decision.



The case to investigate

CASE 4 — Button Clue: a press becomes an on/off value. Meaning: a human event becomes **data**.

Discuss what could be confused

- Which two pictures are easiest to confuse? What **evidence** separates them?

Do not give the sensor an extra power

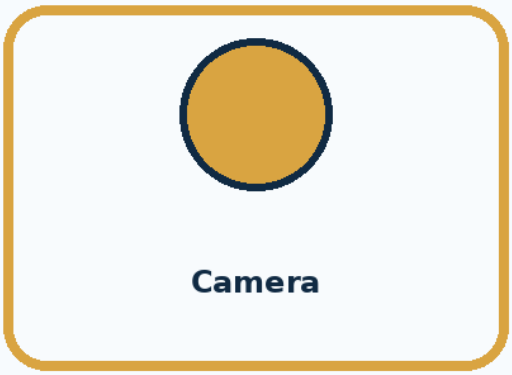
BOUNDARY Do not say a **sensor** sees, hears, or understands. It measures and sends a representation.

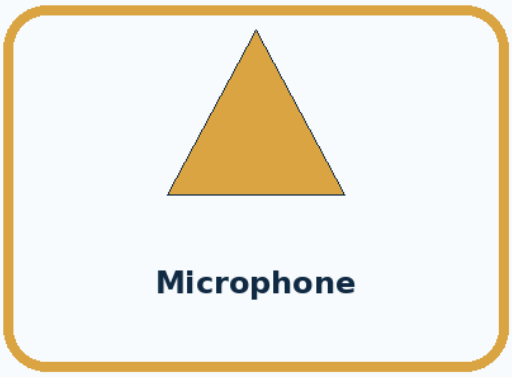
▼ Earlier wording, alternative practice and picture transcripts

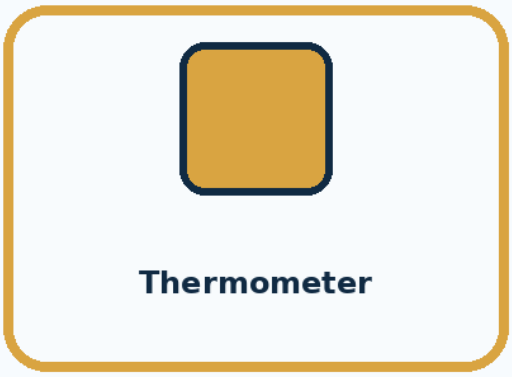
PICTURE GALLERY

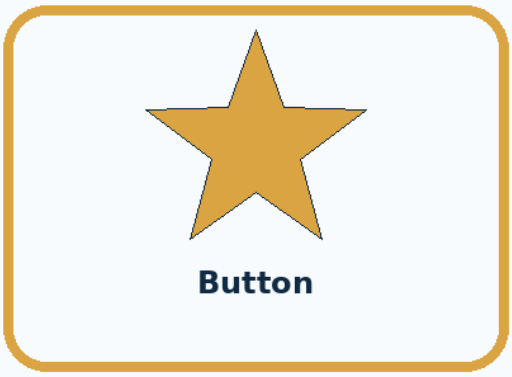
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Camera Clue: light becomes pixel values. Meaning: **measurement**, not meaning.

CASE 2 — Microphone Clue: air-pressure changes become samples. Meaning: **measurement**, not understanding.

CASE 3 — Thermometer Clue: heat becomes a temperature number. Meaning: **measurement**, not a weather decision.

CASE 4 — Button Clue: a press becomes an on/off value. Meaning: a human event becomes **data**.

Change one thing in the pictured case

Change: Move only B into brighter light.

Prediction: The next gap could change for reasons other than a difference in device accuracy.

Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

MAIN LESSON · SECTION 6

A repeated gap does not choose the accurate device

> Compare the same quantity at matching times and locations, then limit your claim.

Given facts: two fictional light sensors

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Neither device has been compared with a trusted reference.

Read the paired measurements

AI ACADEMY / AI-1/3

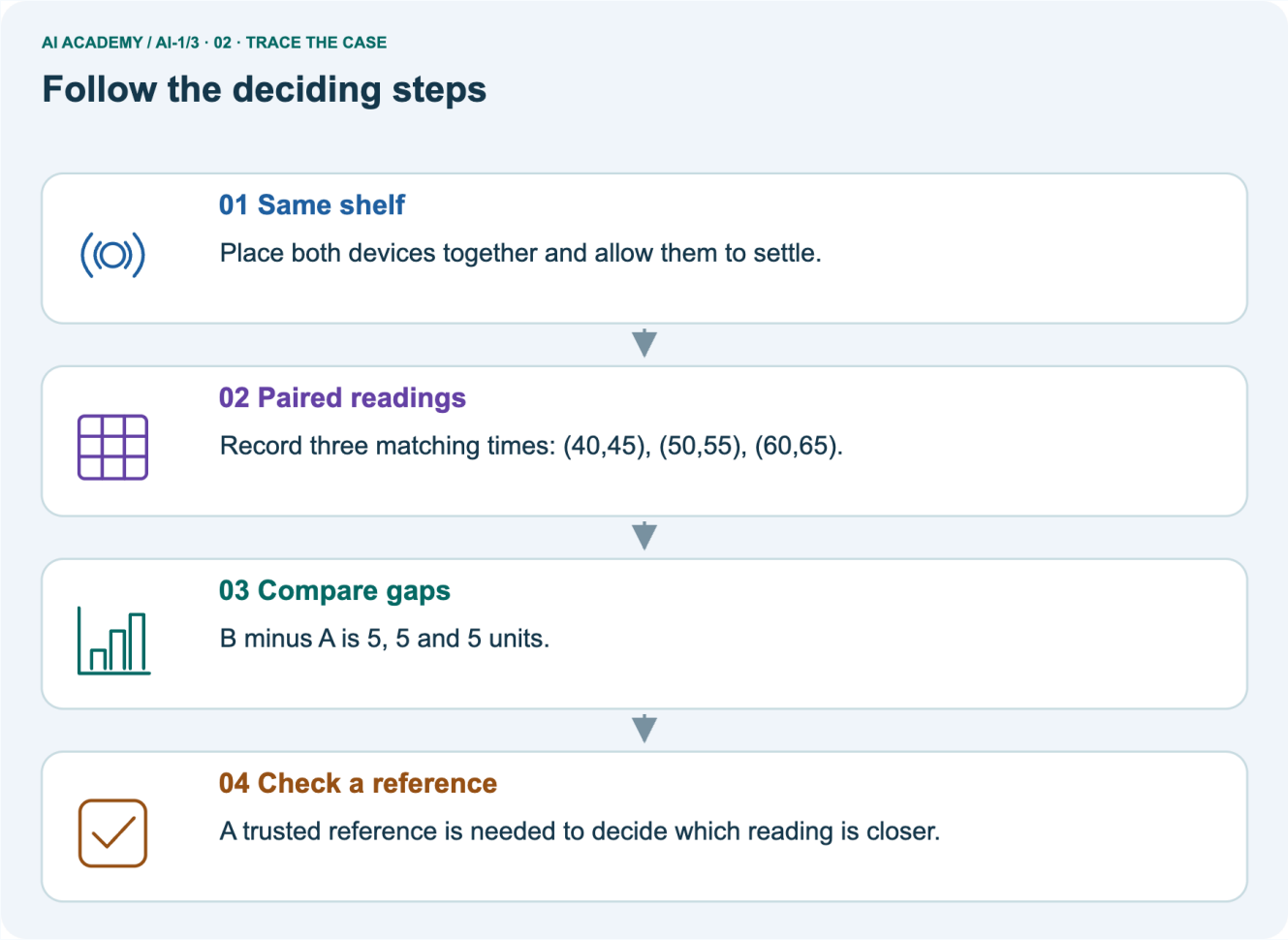
Simultaneous light readings

	Sensor A	Sensor B	B minus A
Time 1	40	45	5
Time 2	50	55	5
Time 3	60	65	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the fair comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

What the repeated gap supports

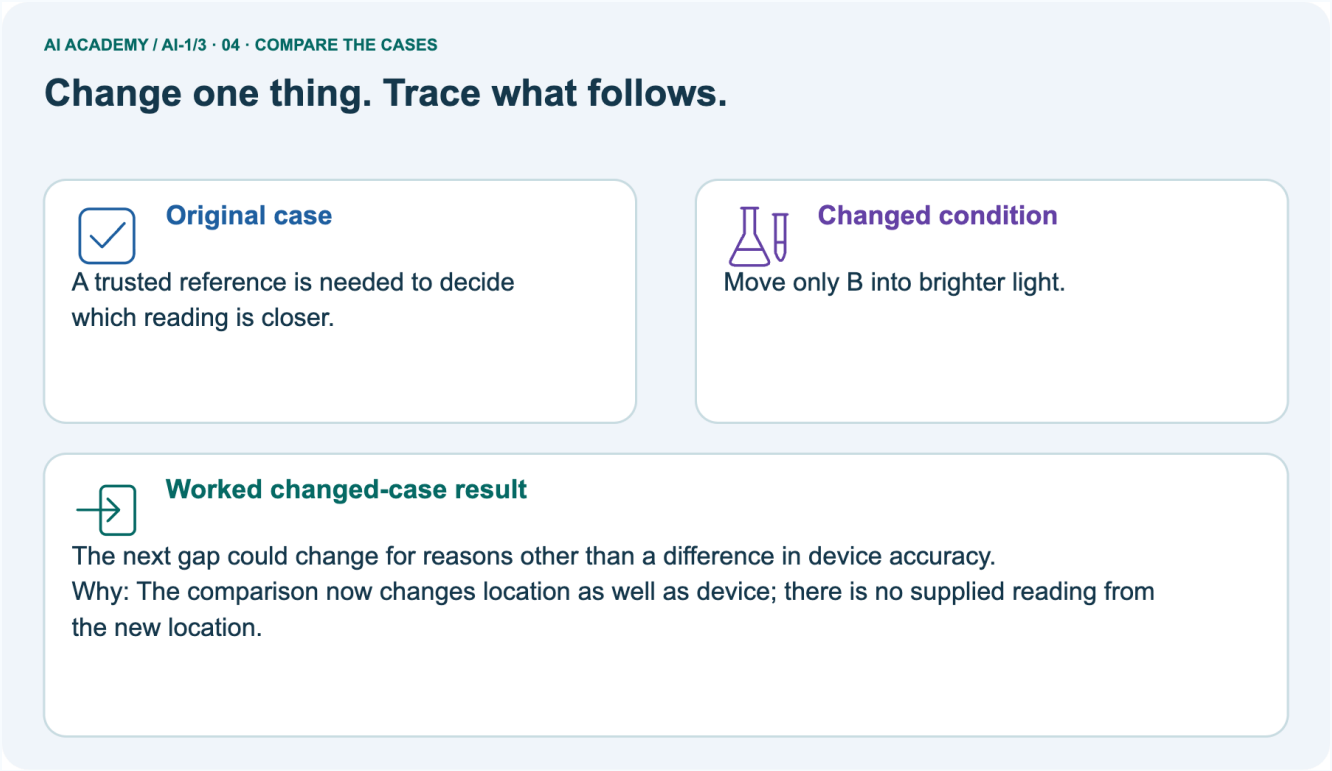
A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently.

Predict after moving only B

- Before checking: move only B into brighter light. Can the earlier gap of 5 tell you its exact new reading? Explain which comparison condition changed and what information is missing.

▼ Check the changed comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

What three pairs cannot establish

Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

Explain the missing information

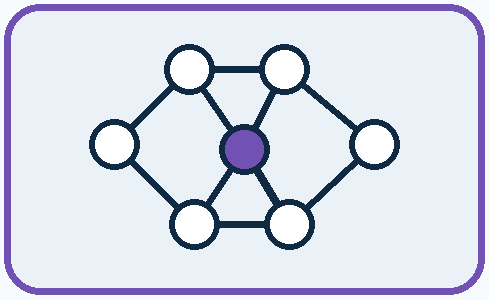
- Read the labelled picture: Why can we calculate the earlier gap but not predict an exact new B reading?

▼ Earlier wording, alternative practice and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A classroom weather station measures light, sound, and temperature but does not decide what those **measurement**s mean.

STEP 1 — OBSERVE light, sound, or heat from the environment

STEP 2 — TRACE a **sensor** converts the **signal** into recorded values

STEP 3 — CONCLUDE numbers or pixel/sound **data** sent to a later **process**

MODEL REASONING 820 could be a light **measurement**, 24 C comes from a temperature **sensor**, and changing sound values come from a microphone.

The numbers alone do not prove whether the plant needs water or whether a voice was present; later **process**ing must interpret them.

DO NOT OVERCLAIM Do not say a **sensor** sees, hears, or understands. It measures and sends a representation.

What new **evidence** could change this answer?

A closer explanation

Two devices are placed side by side in a shaded room and allowed to settle. At three agreed times, device A reads 20, 21 and 20 degrees C; device B reads 22, 23 and 22 degrees C. B is 2 degrees higher on every comparison.

That pattern is **evidence** of a consistent difference under these test conditions.

Time	A in degrees C	B in degrees C	B minus A
10:00	20	22	2
10:05	21	23	2
10:10	20	22	2

We cannot conclude from agreement patterns alone that A is correct and B is wrong. A suitable checked reference can help investigate.

We also need to consider whether the devices have different response times, resolutions or placement. Three comparisons in one room do not prove performance in every temperature range.

Calibration is a reference check, sometimes followed by an adjustment. Do not simply subtract 2 from every future B reading because these three pairs happened to differ by 2.

A correction needs support over the conditions where it will be used.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sensors Are Messengers

The facts we will use

AI ACADEMY / AI-1/3

Simultaneous light readings

	Sensor A	Sensor B	B minus A
Time 1	40	45	5
Time 2	50	55	5
Time 3	60	65	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/3

Sensors Are Messengers

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Same shelf

Place both devices together and allow them to settle.

2. Paired readings

Record three matching times: (40,45), (50,55), (60,65).

3. Compare gaps

B minus A is 5, 5 and 5 units.

4. Check a reference

A trusted reference is needed to decide which reading is closer.

CHANGE ONE THING
Move only B into brighter light.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Same shelf	Place both devices together and allow them to settle.
2. Paired readings	Record three matching times: (40,45), (50,55), (60,65).
3. Compare gaps	B minus A is 5, 5 and 5 units.
4. Check a reference	A trusted reference is needed to decide which reading is closer.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Move only B into brighter light. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Repair a conclusion that outruns its evidence

> A threshold warning does not diagnose everything about a plant.

Find the unsupported step

A warning based on one temperature reading can mean the recorded value crossed a threshold. It cannot automatically mean a plant needs water. That broader claim would need different **evidence**.

When a system makes a surprising decision, check the **unit**s, time, location, **measurement** and rule in order.

Keep the comparison ordinary and safe

For classroom investigations, use teacher-provided records or ordinary safe room conditions. The reasoning task does not require heating devices, opening electronics or testing people.

A clear paper investigation can reveal the same distinction between a reading and an interpretation.

Repair the two claims

- CLAIM A The camera knows there is a cat.
- CLAIM B More **sensor**s always make a system better.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A display says 30 without a unit. State two questions needed before using it in a temperature-warning rule**

▼ **Earlier wording, alternative practice and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Camera	light becomes pixel values	measurement, not meaning
Microphone	air-pressure changes become samples	measurement, not understanding
Thermometer	heat becomes a temperature number	measurement, not a weather decision
Button	a press becomes an on/off value	a human event becomes data

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: hold the right conditions steady

> Separate a change in location from a difference between devices.

Sunny window versus shade

Suppose one classroom device reports 27 degrees C near a sunny window and another reports 22 degrees C in shade. The five-degree difference may reflect the locations. Calling one device faulty skips an alternative explanation.

First write a **prediction**: if location explains the difference, placing both devices together after they settle should reduce it.

A fair comparison changes the suspected factor while keeping other relevant conditions as similar as practical. Record what was changed and what remained uncertain.

Repeated results can strengthen or weaken an explanation, but a short classroom test is not a guarantee about all future **measurement**s.

Trace the boundary yourself

1. Your W2 rule: turn a warning on when recorded temperature is greater than 28 degrees C. Predict the warning for 27, 28 and 29. Explain why a correctly applied rule does not by itself establish an accurate measurement.

2. Use workbook W2 for the connected example "**Sensor**s and **measurement**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Two devices disagree near a window and a shaded wall. Describe a comparison that helps separate location from device differences

Explain, check and switch

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Does stability prove accuracy?

- Use the worked case for **Sensor**s and **measurement**. Proposed explanation: “The more stable device must be accurate”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the changed light reading

Why can we calculate the earlier gap but not predict an exact new B reading?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

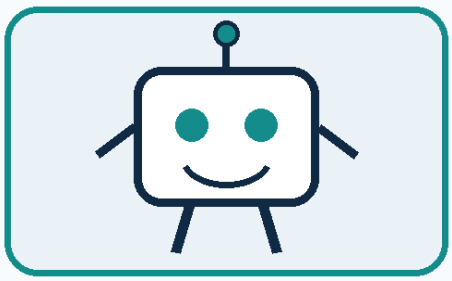
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternative practice and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Sensor**s measure **signal**s; later **process**ing gives those **signal**s meaning.

Use a fresh example to explain **sensor**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: More **sensor**s always make a system better.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Trace a boundary

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate: signal, record or interpretation?

> Use supplied fictional cards to make the sensing boundary visible.

Your three case cards

Use fictional cards instead of recording people: (A) a pattern of light on a camera's detector; (B) changing air pressure at a microphone; (C) the temperature at a thermometer. Match each to a picture record, sound record or temperature record. Keep the physical signal, the recorded data and a later interpretation in three separate columns.

Predict, match and explain

1. Before checking, predict which sensor fits “the room is happy”. The correct boundary is that no card directly measures happiness. For a missing-data case, mark a covered or disconnected sensor’s record as unavailable; do not invent a reading. Swap cards and explain your reasoning to a partner.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep the trace and its limit

Case	Prediction	Observed result	What it means
Normal			
Edge			

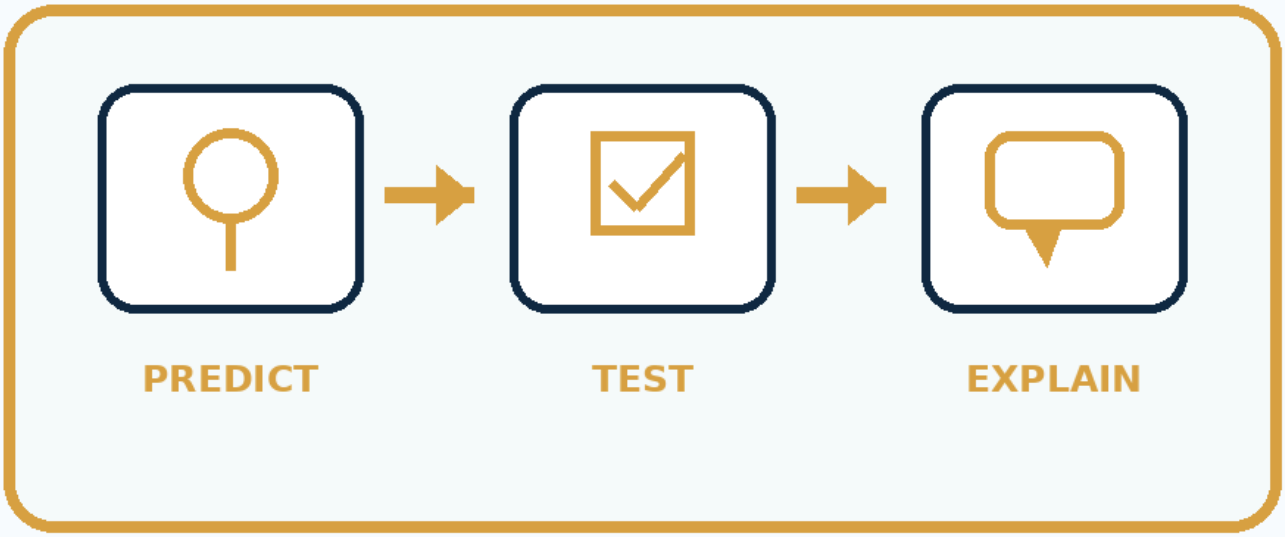
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternative practice and picture transcripts

EXPLORER LAB

Signal Messenger Match

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Match camera, microphone, and temperature **sensor**s to **data**.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: investigate the sound-display disagreement

> Design a fair comparison and explain what two different results would mean.

The new case

A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

My independent explanation, trace and limits

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternative practice and picture transcripts**

Independent case: Investigate a new measurement dispute

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A station reports light level 18 and temperature 22 degrees C. Which **sensor** produced each reading? Can either reading alone tell whether a child feels comfortable? Explain.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Mark which **sensor**, if any, would collect each Classroom Sorter **input**; prefer direct human entry when a **sensor** is unnecessary.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright? Design a fair comparison without using a camera or personal **data**.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Choose whether your sorter needs a sensor

> A direct human entry may serve the classroom need better than adding a device.

This week’s project decision

THIS WEEK'S CHECKPOINT Mark which **sensor**, if any, would collect each Classroom Sorter **input**; prefer direct human entry when a **sensor** is unnecessary.

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions.

Plan the data and human review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep the sensing boundary clear

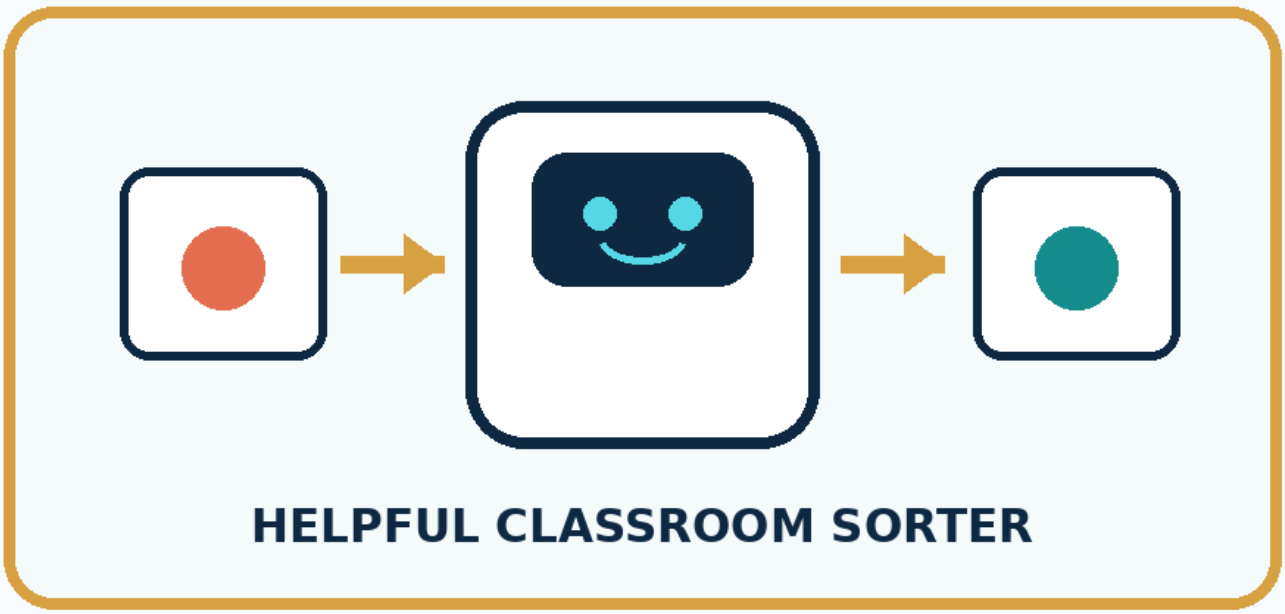
RESPONSIBLE DESIGN Do not say a **sensor** sees, hears, or understands. It measures and sends a representation.

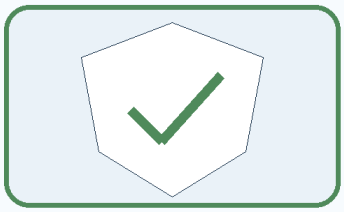
▼ Earlier wording, alternative practice and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: give a measurement, then limit the claim

> Close your notes and explain what a reading supports and what it does not.

Your exit explanation

- Close your notes. Explain this goal in your own words: Describe a **measurement** using its quantity, value, **unit** and conditions

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Choose support and a later return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your measurement word guide**

REFERENCE · SECTION 5

Your measurement word guide

> Use the words to explain a particular reading or comparison.

Core sensor words

Word	Meaning	My example
sensor	device that measures a physical signal	
signal	a changing quantity that can be measured	
measurement	a recorded value describing part of a signal	
data	represented information available for processing	

Useful measurement terms

Word	Meaning
Unit	An agreed amount used to express a quantity, such as a second or a degree Celsius
Variation	Differences among readings or observations
Calibration	Checking and, when appropriate, adjusting a measuring system against a reference

Repair a claim

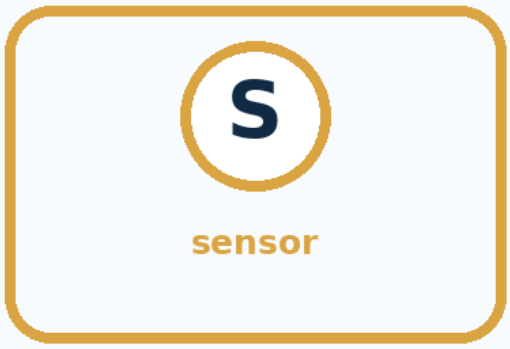
- CLAIM TO REPAIR The camera knows there is a cat.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ **Earlier wording, alternative practice and picture transcripts**

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Pick one next step that improves your explanation.

Choose your challenge

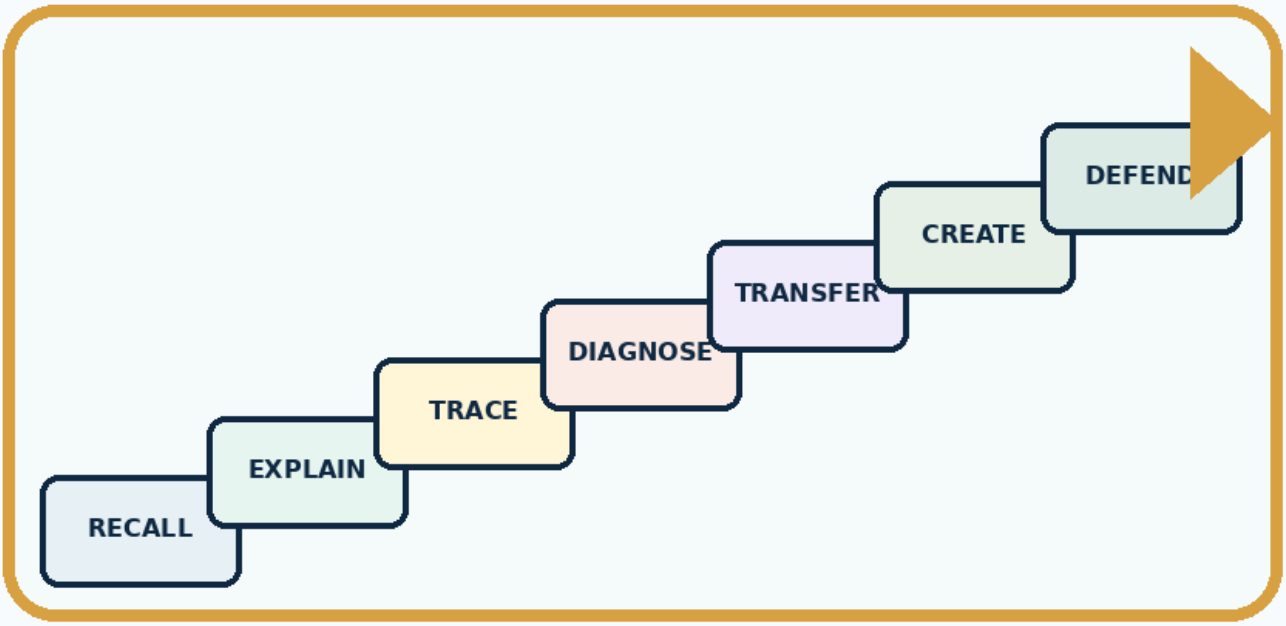
- RECALL Define **sensor**.
- EXPLAIN **Sensor**s measure **signal**s; later **process**ing gives those **signal**s meaning.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: The camera knows there is a cat.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternative practice and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check a measurement explanation**

REFERENCE · SECTION 12

Reference: check a measurement explanation

> Use after practice. The sound-display comparison is this week's selected independent case.

Check your explanation

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternative practice and picture transcripts

MASTERY

Checklist for a measurement explanation

Use this checklist after practice. The sound-display disagreement is this week's selected independent case.



EXPLANATION CHECK Name the quantity and units, trace how the recorded value is used, and explain a fair comparison and its limits.

Independent **evidence**:

SELF-CHECK 820 could be a light **measurement**, 24 C comes from a temperature **sensor**, and changing sound values come from a microphone.

The numbers alone do not prove whether the plant needs water or whether a voice was present; later **process**ing must interpret them.

NEXT BRIDGE Carry today's **evidence** habit into: Rules or Learned Patterns?.

▼ Extra practice: light, sound and feelings

REFERENCE · SECTION 13

Extra practice: light, sound and feelings

> Try the sensor question before opening the explanation.

Read the case and question

A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

Which measures a light-related **signal**? Can either directly measure happiness?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

The light **sensor** measures light; the microphone measures sound. Neither directly measures feelings. **Unit**s and **calibration** matter when comparing devices.

Evidence or stage	Value or result
Cupboard	15
Window	90

▼ Earlier wording, alternative practice and picture transcripts

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Two light readings, one sensor

A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

Which measures a light-related **signal**? Can either directly measure happiness?

The light **sensor** measures light; the microphone measures sound. Neither directly measures feelings. **Unit**s and **calibration** matter when comparing devices.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an extra checklist when it helps you inspect your work.

Targets and evidence

By the end, I can	How I will show it
Describe a measurement using its quantity, value, unit and conditions	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a sensor reading from an interpretation or decision	Explain the deciding step, show a trace or test, and state one limit.
Use repeated comparisons to investigate a measurement problem without inventing certainty	Explain the deciding step, show a trace or test, and state one limit.

Keep the prediction and route

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternative practice and picture transcripts

Start with this question: Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

▼ Extra practice: sensor pictures and feedback

REFERENCE · SECTION 17

Extra practice: sensor pictures and feedback

> Try each question before opening its explanation. Use one paper or browser copy.

First supported question

- Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.
- Which labelled step matches this detail? A trusted reference is needed to decide which reading is closer.
- 1. Compare gaps
- 2. Check a reference
- 3. Paired readings
- 4. Same shelf
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check a reference A trusted reference is needed to decide which reading is closer. A repeated difference of 5 supports investigating an offset between devices.

Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors. Preserve paired readings instead of adjusting them silently.

Changed-condition question

- Changed-case practice: Move only B into brighter light. Which conclusion is supported by the supplied facts?
- 1. The next gap could change for reasons other than a difference in device accuracy.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed condition**

After your attempt, compare with the supported explanation: The next gap could change for reasons other than a difference in device accuracy.

The comparison now changes location as well as device; there is no supplied reading from the new location.

Repair and save

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternative practice and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

</details>

<details>
<summary>Reference: six-picture sensor journey</summary>

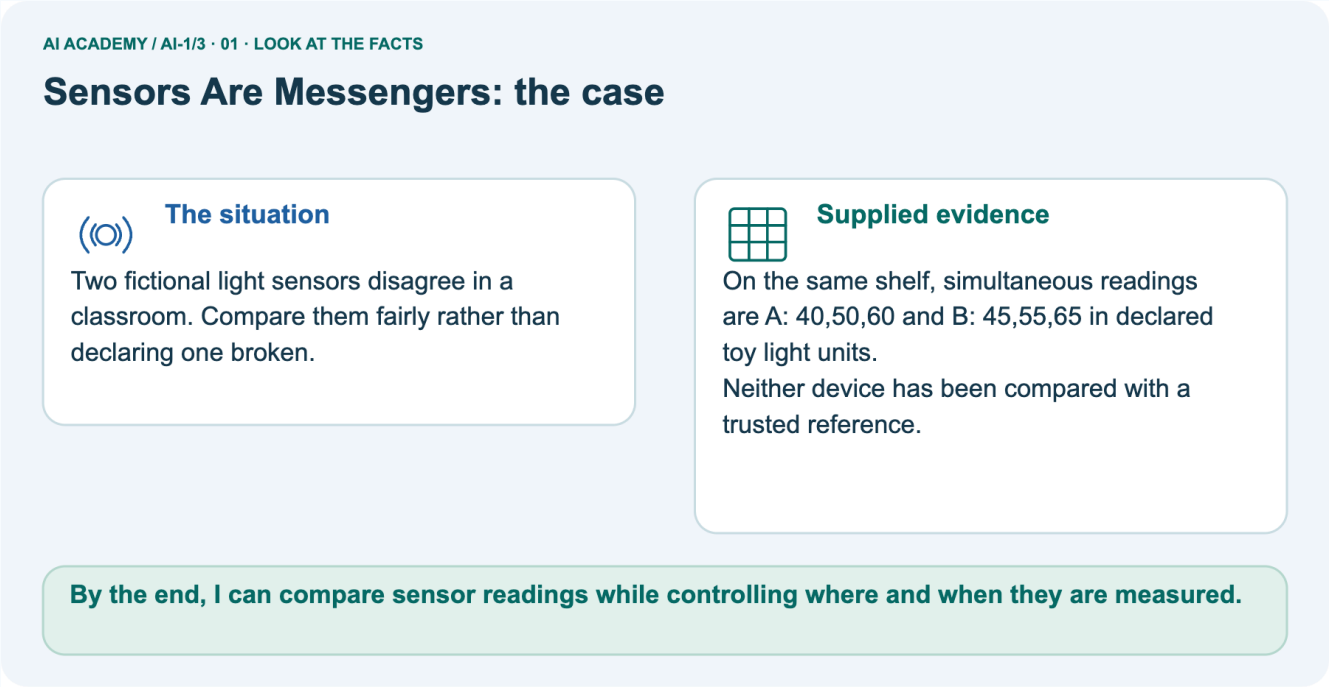
REFERENCE · SECTION 18

Reference: six-picture sensor journey

> Use the image that helps your current question; the main comparison already appears in the lesson.

</details>

01. Sensors Are Messengers: the case



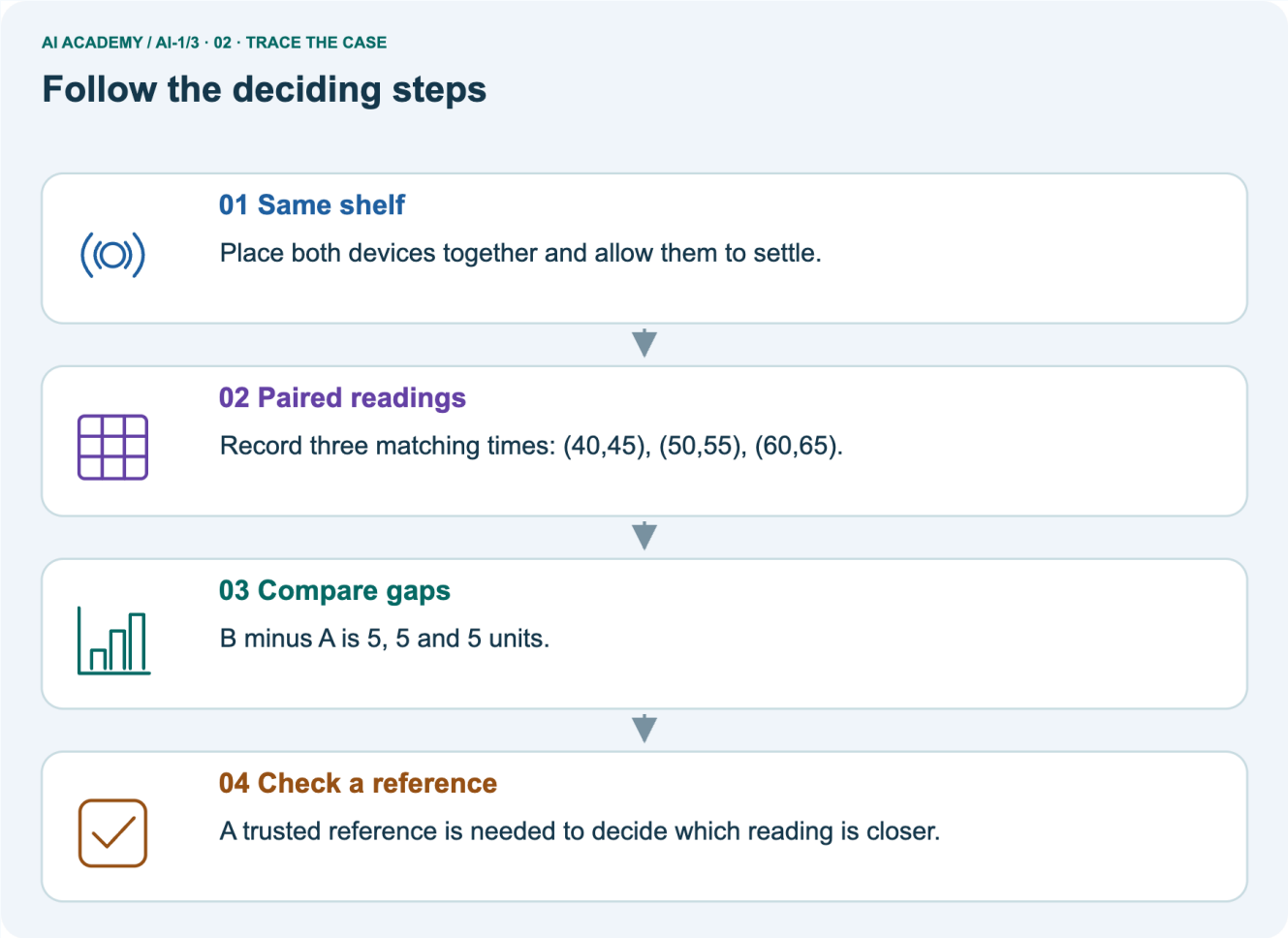
Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken. Supplied facts: On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Neither device has been compared with a trusted reference.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

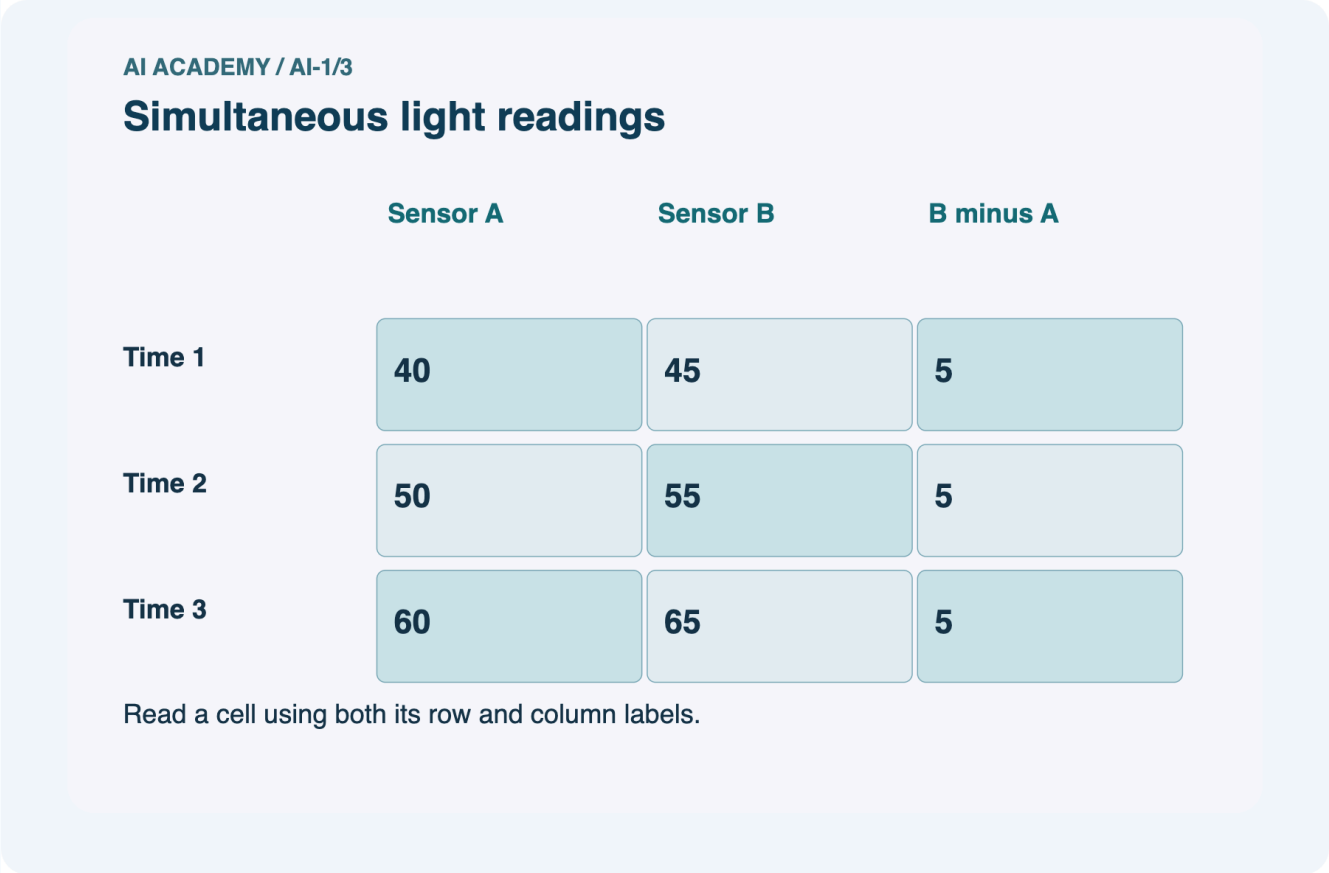


Follow the labelled picture with the matching explanation. Open the vector diagram

Same shelf: Place both devices together and allow them to settle. → Paired readings: Record three matching times: (40,45), (50,55), (60,65). → Compare gaps: B minus A is 5, 5 and 5 **unit**s. → Check a reference: A trusted reference is needed to decide which reading is closer.

Point to each numbered stage and explain which recorded **input** it uses.

03. Simultaneous light readings

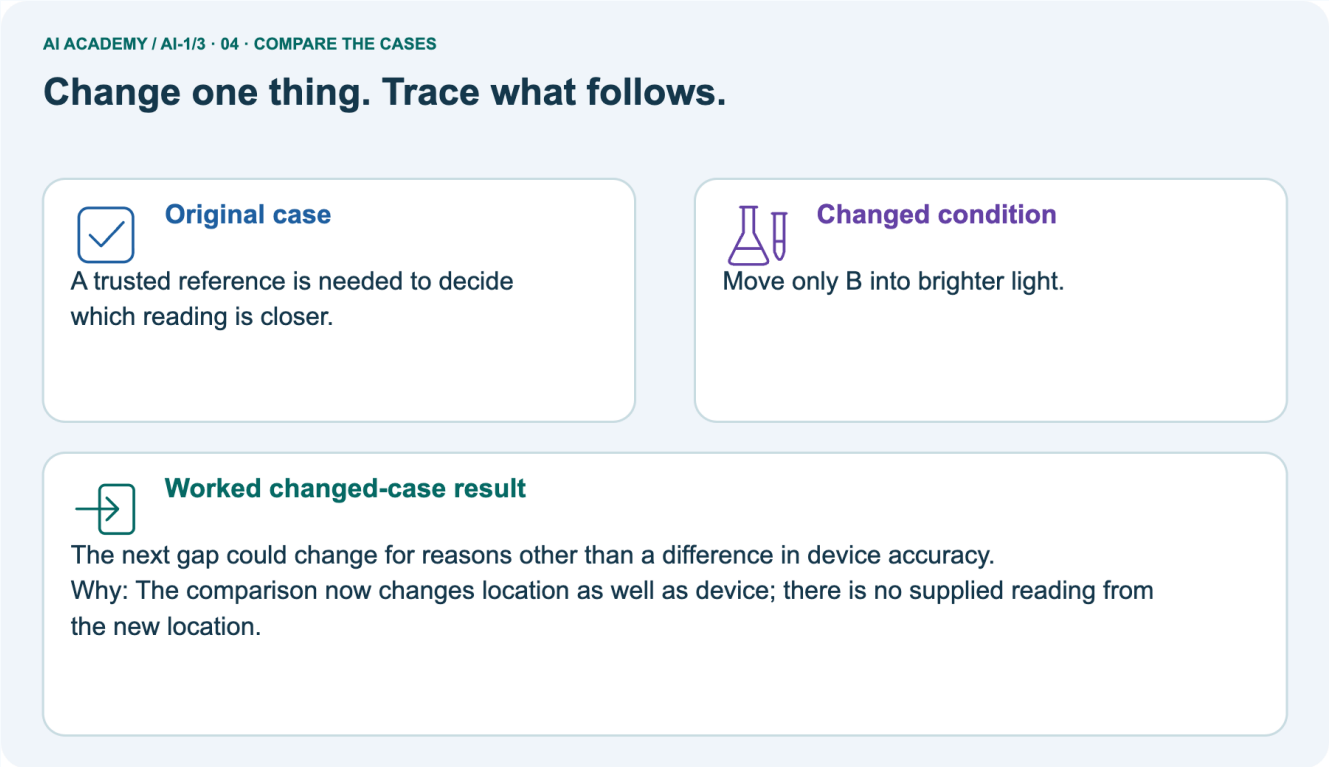


Follow the labelled picture with the matching explanation. Open the vector diagram

Simultaneous light readings. Time 1: **Sensor** A = 40, **Sensor** B = 45, B minus A = 5. Time 2: **Sensor** A = 50, **Sensor** B = 55, B minus A = 5. Time 3: **Sensor** A = 60, **Sensor** B = 65, B minus A = 5.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



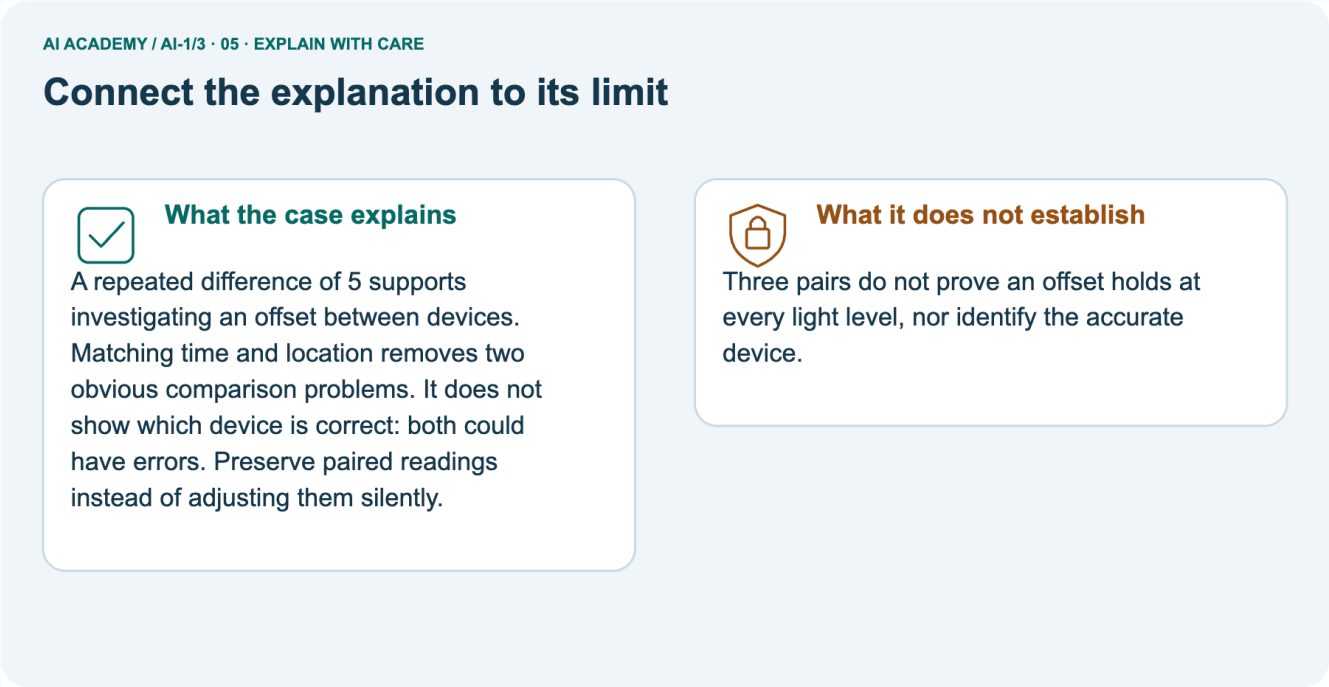
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A trusted reference is needed to decide which reading is closer. Changed condition: Move only B into brighter light.

Expected result: The next gap could change for reasons other than a difference in device accuracy. Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



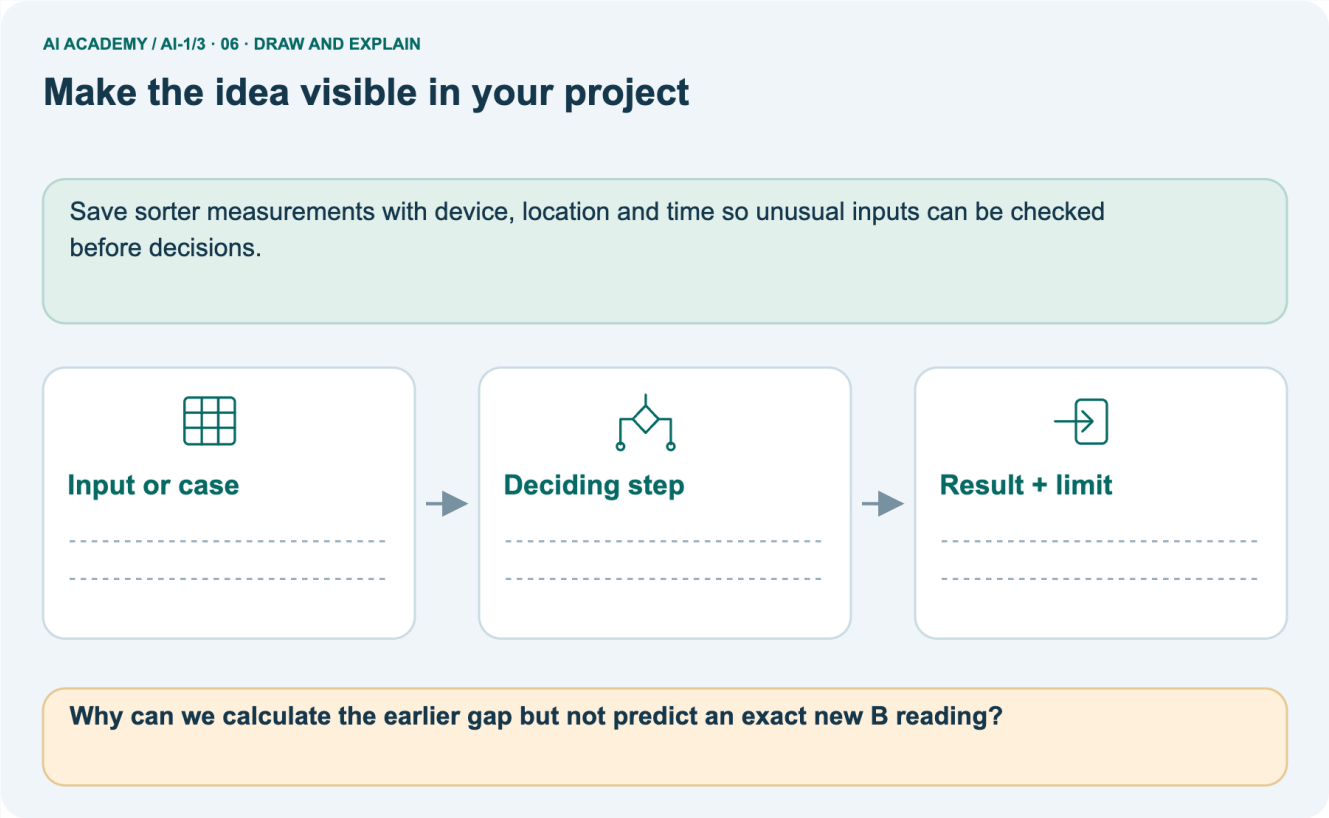
Follow the labelled picture with the matching explanation. Open the vector diagram

A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently. Limit: Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions. Draw your **input**, deciding step, result and limit.

Supported question: Why can we calculate the earlier gap but not predict an exact new B reading?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternative practice and picture transcripts

01. Sensors Are Messengers: the case

02. Follow the deciding steps

03. Simultaneous light readings

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.